

1. Wave function Ψ :

The quantity whose variation make up matter wave is called wave function Ψ .

2. **Matter Wave:** The wave associated with the materialistic particles and represented by Ψ is called the matter wave.

$$\Psi = \Psi(x, y, z, t) \dots \dots \dots (1)$$

The wave function Ψ is dependent upon the position and time as given in equation no (1). Equation (1) represents the value of the wave function associated with moving particle at **position (x,y,z,t)**.

3. Born Interpretation:

The wave function Ψ is not an observable quantity, because the wave function represents the matter wave associated with the particle like electron. Further if we consider an electromagnetic wave or sound waves, the intensity at any point can simply be obtained by squaring the amplitude i.e. $I \propto E_0^2$ where E_0 the amplitude of the electromagnetic wave represented by $E = E_0 e^{-i(\omega t - kx)}$.

In the same way matter wave associated with material particles like electron, represented by wave function Ψ , the square of the amplitude of matter wave (Ψ^2) gives the **probability of finding the particle**. This interpretation was first made by Max Born and called the Born interpretation.

4. Probability of finding the particle:

Probability of finding the particle which is described by wave function Ψ at the point (x,y,z) and at time t is proportional to the value $|\Psi|^2$.

Usually the wave function Ψ is represented by

$$\Psi = A e^{-i(\omega t - kx)} \dots \dots \dots (2)$$

The wave function ψ is usually a complex quantity having real and imaginary parts. $\left(\omega = \frac{E}{\hbar} \text{ sec}^{-1}\right)$ is the angular frequency of the associated wave and $\left(k = \frac{p}{\hbar} \text{ m}^{-1}\right)$ is the wavenumber. E is the energy and p is the momentum of the associated wave with the particle.

$$\text{So the probability of finding the particle is } P = |\Psi|^2 \dots \dots \dots (3)$$

Probability of finding the particle in the range (x_1 to x_2) can be written as

$$P_{x_1, x_2} = \int_{x_1}^{x_2} |\Psi|^2 dx = \int_{x_1}^{x_2} \Psi \Psi^* dx \dots \dots \dots (4)$$

Where $\Psi^* = A e^{i(\omega t - kx)}$ complex conjugate of the wave function

$$\Psi = A e^{-i(\omega t - kx)} \text{ represented by equation (2)}$$

5. Normalization:

Suppose wave function ψ is represented by

$$\Psi = A e^{-i(\omega t - kx)} \dots \dots \dots (5)$$

complex conjugate of the wave function represented by equation (5)

$$\Psi^* = A e^{i(\omega t - kx)} \dots \dots \dots (6)$$

Therefore a wave function is said to be normalized if

$$\int_{-\infty}^{\infty} |\Psi|^2 dx = \int_{-\infty}^{\infty} \Psi \Psi^* dx = 1 \dots \dots \dots (7)$$

From equation (3) equation (7) can be written as

$$\text{or } \int_{-\infty}^{\infty} P dx = 1 \dots \dots \dots (8)$$

Thus Equation (7) or (8) represents the normalization condition.

6. Properties of wave function Ψ :

- (a) Ψ must be continuous everywhere.
- (b) Ψ must be single valued
- (c) $\frac{\partial \Psi}{\partial x}$, $\frac{\partial \Psi}{\partial y}$, and $\frac{\partial \Psi}{\partial z}$ must be continuous and single values everywhere.
- (d) ψ must be normalizable which means that
 $\Psi \rightarrow 0$ and $x \rightarrow \pm\infty$, $y \rightarrow \pm\infty$ and $z \rightarrow \pm\infty$ in order that
 $\int_{-\infty}^{\infty} |\Psi|^2 dx = \int_{-\infty}^{\infty} \Psi \Psi^* dx = 1$

7. Schrodinger's Equation:

Schrodinger's equation is fundamental principle of Quantum Mechanics: this cannot be derived from any other basic principles of physics.

(a) Time dependent Schrodinger Equation:

Suppose a wave function Ψ associated with a particle moving freely in $+x$, direction is given by

$$\Psi = Ae^{-i(\omega t - kx)} \dots \dots \dots (1)$$

Where $\left(\omega = \frac{E}{\hbar} \text{ sec}^{-1}\right)$ is the angular frequency and $\left(k = \frac{p}{\hbar} \text{ m}^{-1}\right)$ is the wave number of the associated wave with the particle.

So wave function can be written as, after replacing the angular frequency and wavenumber

$$\Psi = Ae^{-i\left(\frac{Et}{\hbar} - \frac{px}{\hbar}\right)}$$

$$\Psi = Ae^{-\frac{i}{\hbar}(Et - px)} \dots \dots \dots (2)$$

Differentiating Equation (2) twice with respect to x , we have

$$\frac{d\Psi}{dx} = A \frac{ip}{\hbar} e^{-\frac{i}{\hbar}(Et - px)}$$

$$\frac{d^2\Psi}{dx^2} = A \left(\frac{ip}{\hbar}\right)^2 e^{-\frac{i}{\hbar}(Et - px)}$$

$$\frac{d^2\Psi}{dx^2} = -\left(\frac{p}{\hbar}\right)^2 A e^{-\frac{i}{\hbar}(Et - px)}$$

$$\frac{d^2\Psi}{dx^2} = -\frac{p^2\Psi}{\hbar^2} \dots\dots\dots(3)$$

$$\Rightarrow p^2\Psi = -\hbar^2 \frac{d^2\Psi}{dx^2} \dots\dots\dots(4)$$

Again differentiating equation (2) with respect to t , we have

$$\frac{d\Psi}{dt} = A \left(-\frac{iE}{\hbar} \right) e^{-\frac{i}{\hbar}(Et-px)}$$

$$\frac{d\Psi}{dt} = \left(-\frac{iE\Psi}{\hbar} \right) \dots\dots\dots(5)$$

$$\Rightarrow E\Psi = i\hbar \frac{d\Psi}{dt} \dots\dots\dots(6)$$

Total energy E , of the moving particle can be written as,

Total energy of the particle = Kinetic energy + Potential energy

$$E = \frac{p^2}{2m} + V(x, t) \dots\dots\dots(7)$$

Multiplying both the sides by ψ in equation (7), we have

$$E\Psi = \frac{p^2\Psi}{2m} + V(x, t)\Psi \dots\dots\dots(8)$$

Now substituting the value of $E\psi$ and $p^2\psi$ in equation (8) we have

$$i\hbar \frac{d\Psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} + V\Psi \dots\dots\dots (9)$$

Equation (9) represents the time dependent Schrodinger Equation

For three dimensions, Equation (9) can be written as

$$i\hbar \frac{d\Psi}{dt} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2} + \frac{\partial^2\Psi}{\partial z^2} \right) + V\Psi \dots\dots (10)$$

(b) Time independent Schrodinger Equation or (Steady state form of the Schrodinger Equation)

In many situations the potential energy of the particle V , does not depend on time. The forces which acts on it and hence the potential V , vary with position only.

In this case Schrodinger equation can be simplified by removing the time t dependence.

So the wave function Ψ can be written as:

$$\Psi = Ae^{-\frac{i}{\hbar}(Et - px)} = Ae^{-i\frac{Et}{\hbar}} e^{i\frac{px}{\hbar}}$$

$$\Psi = \psi e^{-i\frac{Et}{\hbar}} \dots\dots\dots(1)$$

Where Ψ is the product of time dependent function, $e^{-i\frac{Et}{\hbar}}$, and position dependent function $\psi = Ae^{i\frac{px}{\hbar}}$.

We have already discussed the time dependent form of the Schrodinger equation

$$i\hbar \frac{d\Psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} + V\Psi \dots\dots\dots(2)$$

Substituting the value of equation (1) in (2) we have

$$i\hbar \frac{d\psi e^{-i\frac{Et}{\hbar}}}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi e^{-i\frac{Et}{\hbar}}}{dx^2} + V\psi e^{-i\frac{Et}{\hbar}}$$

$$\Rightarrow i\hbar \psi \frac{d e^{-i\frac{Et}{\hbar}}}{dt} = -\frac{\hbar^2}{2m} e^{-i\frac{Et}{\hbar}} \frac{d^2\psi}{dx^2} + V\psi e^{-i\frac{Et}{\hbar}}$$

$$\Rightarrow i\hbar \left(-\frac{iE}{\hbar}\right) \psi e^{-i\frac{Et}{\hbar}} = -\frac{\hbar^2}{2m} e^{-i\frac{Et}{\hbar}} \frac{d^2\psi}{dx^2} + V\psi e^{-i\frac{Et}{\hbar}}$$

$$\Rightarrow E \psi = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi$$

$$\Rightarrow \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0 \dots\dots\dots (3)$$

Equation (3) is known as time independent Schrodinger Equation for one dimension.

For three dimensions, Equation (3) can be written as

$$\left(\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} \right) + \frac{2m}{\hbar^2} (E - V) \Psi = 0 \dots\dots\dots(4)$$